Choosing a Correctly Sized Analysis for the Biomarker-by-Drug Interaction in a Hybrid N-of-1 Trial

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Purpose. This paper chooses a primary analysis, and a set of sensitivity analyses, for the biomarker-by-drug interaction in a Hybrid N-of-1 trial of prazosin with blood pressure as the candidate predictive biomarker. The criterion is that the test must hold its nominal size before power is considered.

The evidence is the covariance study (`08-covariance-study.R`), run in three passes on the same simulated trials: 63 analyses, 16 cells, and 378,000 model fits, with one failed fit. It also uses a separate check of the drug-exposure coding used in manuscript 02 (`10-dbc-preexposure-check.R`). The paper consolidates the discussion in `docs/40` to `docs/43` and corrects several statements made there before the results were in.

Status note (2026-10-07). Two parts of this paper are superseded. (1) The primary recommendation (`us-KR` with $D_{bc,it}$, Sections 1 and 9) is replaced by analysis F1 of the exploratory stress test (`docs/46`, Amendments 1 and 2; `docs/49`, Section 7). In that test the unstructured Kenward-Roger model (analysis A) rejected 0.057 pooled over the 11 standard null cells and 0.087 at $N = 35$, failing the strict size criterion, and rejected 0.195 to 0.416 in the three natural-course and placebo prognostic null cells (verified, `decision-screen-ds-nosym-all14.csv`). (2) Section 4.4 understates the flexible-time size excess; see the note there.

(Updated 2026-10-09: the pre-exposure coding check (`10-dbc-preexposure-check.R`, the table of the section on manuscript 02's coding) now uses anchored carryover at $t_{1/2} > 0$; see `docs/53`. Values changed by at most 0.004; conclusions unchanged. The covariance-study results elsewhere in this paper still come from the run with the published recursive carryover; its anchored rerun is pending and they have not been changed.)

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here or in `docs/41`.

Table 1: Notation index and glossary of symbols, meanings and status

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t (code `Sx`); $t = 0$ is baseline	canonical
BL_i	baseline symptom score	canonical
B_i	biomarker (code `bm`); centered at its sample mean in every analysis	canonical
D_{it}	binary drug state (code `Db`); 0 at baseline	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`): 1 on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after exposure, 0 before any exposure	canonical; the pre-exposure value is made explicit here
t_{sd}	time since discontinuation, in weeks; 0 before first exposure	canonical
$t_{1/2}$	carryover half-life of the data-generating process, in weeks	canonical
$t^a_{1/2}$	half-life assumed by the analysis when it builds $D_{bc,it}$	local
λ	decay rate, $\ln 2/t_{1/2}$	canonical
c_{bm}	covariance-moderation strength: on-drug correlation of the biomarker with the drug response	canonical
ρ	AR(1) serial correlation per week	canonical
β_B	biomarker main effect	local (`docs/41`)
β_D, $\beta_{bm:D}$	drug main effect; biomarker-by-drug interaction, the estimand	canonical
α	nominal level, 0.05	canonical
κ	mean model-based standard error over the empirical standard deviation of the estimate; below 1 means anticonservative	canonical
ν	denominator degrees of freedom	canonical
Σ	covariance matrix of a participant's outcomes	local
ε_{it}	residual of the analysis model	canonical
β_0, β_t	intercept; coefficient of linear time (`docs/40`)	local

Symbol	Meaning	Status
w_t	calendar week of visit t (code `t`)	local (`docs/40`)
X_{it}	exposure regressor of the analysis, D_{it} or $D_{bc,it}$	local
$f(t)$	time adjustment: linear, spline or visit factor	local
$\mathbf{1}\{t = 0\}$	indicator of the baseline row	local
N	total sample size, 70	canonical

Sign convention. The response components reduce symptom severity, so β_D and $\beta_{bm:D}$ are negative and c_{bm} is positive.

Analysis labels. Each analysis is named by its baseline coding, working covariance and inference, for example `BL+us-KR`.

- **Baseline coding.** No prefix: baseline is a ninth row of the response, coded off drug at $t = 0$ (the published coding; `docs/41` calls it the response coding). `BL+`: baseline is a covariate and the eight post-baseline visits are the response.
- **Working covariance.**
  - `cs`: compound symmetry; with a row-coded baseline and linear time this is the published random-intercept model.
  - `csh`: heterogeneous compound symmetry.
  - `us`: unstructured; 45 parameters over 9 rows, 36 over 8.
  - `sp_exp`: exponential spatial correlation in continuous time.
  - `RI+CAR1`: a random intercept plus `corCAR1` residuals, fitted with `nlme`. This is manuscript 02's analysis, G6 with D_{it} and G8 with $D_{bc,it}$.
- **Inference.**
  - `Satt`: asymptotic model-based covariance of the estimates, with Satterthwaite ν.
  - `KR`: Kenward-Roger [2], with the `Kenward-Roger-Linear` covariance for `us`.
  - `CR2`: the bias-reduced cluster-robust (sandwich) covariance, with Satterthwaite ν (Bell and McCaffrey [3]).
- **Suffixes.** `-Dbc0.5`, `-Dbc1` and `-Dbc2` replace D_{it} with $D_{bc,it}$ at $t^a_{1/2} = 0.5$, 1 and 2 weeks. `-sepbm` frees the biomarker main effect at baseline (Section 6).
- **Time adjustment.** Linear t unless stated; `spline` is a natural spline with 3 degrees of freedom, and `visit` is a visit factor.

Glossary.

- **Size.** The rejection rate of the interaction test when $\beta_{bm:D} = 0$.
- **Correctly sized.** Defined in Section 2.
- **Size-adjusted power.** Power when each analysis rejects at the 5% quantile of its own null p-values in the matching null cell, so that analyses are compared at a common realized size.
- **Construct.** The data-generating process. *CS* is the published compound-symmetry construct with graded coupling at $c_{bm} = 0.25$. *C07* is configuration C (separable AR(1), $\rho = 0.7$) with graded coupling at $c_{bm} = 0.45$ (`docs/36`, `docs/44`). It is the covariance construct of the refined process of paper 01, not the refined process itself: this study kept the published recursive carryover and the fixed published allocation (inspected in `08-covariance-study.R`).
- **Shared biomarker effect.** In the row coding, a single β_B describes the biomarker's association with the outcome at baseline and at every post-baseline off-drug visit.
- **Constancy assumption.** The assumption that this association is in fact the same at baseline and at post-baseline off-drug visits (Section 6).

1. Summary

1. **The working covariance and the small-sample correction determine the size.** The time adjustment, the baseline coding and the exposure coding move the mean Type I error by at most about 0.01 (Section 4).
2. **Kenward-Roger and CR2 repair different failures, so neither substitutes for the other.**
   - Kenward-Roger corrects for the cost of estimating a covariance that is correctly specified. With `us` it brings the mean size from 0.090 to 0.058. With a misspecified `cs` or `csh` it does nothing: 0.15 to 0.16 in CO under AR(1) data.
   - CR2 corrects for misspecification of a parsimonious working covariance. With `cs`, `csh`, `sp_exp` or `RI+CAR1` it gives a mean size of 0.044 to 0.054 in every construct. With `us` it fails: 0.086, no better than no correction.
3. **Correctly sized in every one of the eight null cells (Section 4):**
   - nominal: `cs-CR2`, `csh-CR2`, `sp_exp-CR2`, `RI+CAR1-CR2` and `BL+cs-CR2`, mean 0.044 to 0.054;
   - slightly liberal: `us-KR` and `BL+us-KR`, mean 0.058 to 0.059, no cell above 0.072.

   The published analysis (`cs-Satt`) is correctly sized under CS data and in Hybrid. Under AR(1) data in CO it rejects 0.15 to 0.16 of the time.
4. **At a common size, power differs little under CS data.** In Hybrid without carryover, every correctly sized analysis except `sp_exp` and

`RI+CAR1` has size-adjusted power of 0.66 to 0.72. Under AR(1) data the row-coded `us`, `RI+CAR1` and `sp_exp` analyses reach 0.94. The correctly sized baseline-covariate analyses reach 0.76 to 0.78 (Section 5).

5. **The row coding's advantage under AR(1) comes from the shared biomarker effect, not from the covariance.** Freeing β_B at baseline (`us-KR-sepbm`) gives the same power as baseline as a covariate: 0.780 against 0.792 in Hybrid. The advantage is valid only under the constancy assumption, which the null cells cannot test because the construct satisfies it by design (Section 6).
6. $D_{bc,it}$ **at a short assumed half-life is a near-free insurance against carryover.** With `us-KR` in Hybrid at $t^a_{1/2} = 0.5$:

   - without carryover it costs at most 0.016 in power;
   - at $t_{1/2} = 1$ week it gains 0.04 to 0.06;
   - its size is unchanged.

   A longer assumed half-life gains more under long carryover but costs up to 0.14 without it. With $t^a_{1/2} = t_{1/2}$ the estimate recovers the no-carryover interaction (Section 7).
7. **Manuscript 02's** $D_{bc,it}$ **codes off-drug visits before the first exposure as 1.** In CO this makes the indicator constant for every participant randomized to placebo first. On the same data, the corrected coding restores power from 0.484 to 0.744 under CS and from 0.256 to 0.556 under AR(1) (verified, Section 8). This is the likely cause of manuscript 02's finding that the exposure-weighted analysis collapses under CO.
8. **Recommendation (Section 9; superseded 2026-10-07 by `docs/46` and `docs/49`, which recommend F1).**

   - **Primary analysis:** an MMRM with unstructured covariance and Kenward-Roger inference, baseline as a response row, linear time, and $D_{bc,it}$ at a fixed $t^a_{1/2} = 0.5$ week.
   - **Sensitivity analyses:** the same model with β_B freed at baseline (equivalently, baseline as a covariate), and `RI+CAR1-CR2`.
   - **If the protocol cannot defend the constancy assumption,** the freed model should be primary.

2. The question and the criterion

A trial's primary test of a predictive biomarker is useful only if its Type I error is what the protocol states. In this setting, an analysis that rejects 8% of the time under the null cannot be rescued by its power, so the analyses are first screened on size, and power is compared only among those that pass.

Monte Carlo precision. Each null cell has 500 replicates, so the Monte Carlo standard error (MCSE) of a rejection rate of 0.05 is 0.0097. A single cell between 0.031 and 0.069 is consistent with nominal size. With 8 null cells, the pooled rate rests on 4,000 replicates (MCSE 0.0034), with a 95% band of 0.043 to 0.057.

Criterion. An analysis is called:

- **nominal** if its mean size over the eight null cells is within 0.043 to 0.057 and no cell exceeds 0.075;
- **slightly liberal** if the mean is 0.057 to 0.065 and no cell exceeds 0.075;
- **conservative** if the mean is below 0.043;
- **liberal** otherwise.

An analysis is *correctly sized* if it is nominal or slightly liberal. The upper cell limit of 0.075 is 2.6 MCSE above nominal. With 63 analyses and 8 cells, a few individual cells above 0.069 are expected by chance, so the cell limit is set above the single-cell band.

3. The study

Constructs. Both constructs use the published means and standard deviations of `04-power-simulation.R` (which reproduces the `58b32a9` matrices), $N = 70$, and the published allocation (18/18/17/17 in Hybrid, 35/35 in CO). Both use graded coupling: full coupling on drug, $c_{bm} 2^{-t_{sd}/t_{1/2}}$ off drug after exposure, zero before exposure (`docs/37`, Appendix A).

- **CS:** compound symmetry within each response component (0.8 at every lag), cross-component correlations 0.2 and 0.1, $c_{bm} = 0.25$ (below the ceiling of 0.256).
- **C07:** separable AR(1) at $\rho = 0.7$ per week, $c_{bm} = 0.45$ (below the ceiling of 0.480 in Hybrid and 0.616 in CO).

Designs.

- **Hybrid:** visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 after baseline, with four paths: discontinuation after week 9 or week 10, crossed with drug-first or placebo-first crossover.
- **CO:** eight visits 2.5 weeks apart, with drug-first and placebo-first paths.

Cells. 2 constructs × 2 designs × $t_{1/2} \in \{0, 1\}$ week × {alternative, null}, 16 cells. The null twin of each cell sets $c_{bm} = 0$. Alternative cells have 250 replicates (power MCSE at most 0.032); null cells have 500.

Analyses. The mean model is $\beta_0 + \beta_B B_i + \beta_D X_{it} + f(t) + \beta_{bm:D} B_i X_{it}$, with X_{it} either D_{it} or $D_{bc,it}$ and $f(t)$ linear, spline or visit factor. The baseline coding, working covariance and inference vary as in the glossary. All analyses are fitted with `mmrm` [4], except `RI+CAR1` (`nlme` plus `clubSandwich`).

Three passes, paired.

Table 2: Covariance study passes with analyses, model fits and failed fits

Pass	Analyses	Fits	Failed fits
Main	`cs`, `csh`, `us`, `sp_exp` with Satterthwaite; `us-KR`; `BL+cs`, `BL+us` with Satterthwaite; each time form	120,000	0
Follow-up	KR and CR2 versions of the above; `us-KR-sepbm`	138,000	0
Exposure	$D_{bc,it}$ at three $t^a_{1/2}$ under `us-KR`, `us-CR2` and `RI+CAR1-CR2`; D_{it} under `RI+CAR1-CR2`	120,000	1

Every pass uses the same per-chunk seeds, so the analyses are fitted to identical simulated trials. Differences between analyses are therefore paired comparisons, and are more precise than the MCSE of each rate suggests.

4. Size

4.1 The size table

Mean, minimum and maximum size over the eight null cells (linear time), with the maximum in each group of cells: the four CS cells, the two C07 Hybrid cells and the two C07 CO cells. The last numeric column is the smallest κ over the eight cells.

Table 3: Null size and minimum κ by analysis over the eight null cells

Analysis	Mean	Min	Max, CS	Max, C07 Hybrid	Max, C07 CO	Min κ	Class
`cs-Satt`	0.079	0.042	0.058	0.066	0.158	0.73	liberal
`cs-KR`	0.079	0.042	0.058	0.066	0.158	0.73	liberal
`csh-KR`	0.073	0.040	0.058	0.048	0.154	0.75	liberal
`cs-CR2`	0.051	0.042	0.064	0.044	0.060	0.95	nominal
`csh-CR2`	0.054	0.048	0.064	0.054	0.056	0.93	nominal
`sp_exp-Satt`	0.042	0.028	0.062	0.048	0.034	0.95	conservative
`sp_exp-CR2`	0.048	0.040	0.064	0.050	0.044	0.92	nominal
`RI+CAR1-CR2`	0.051	0.040	0.064	0.056	0.050	0.94	nominal
`us-Satt`	0.090	0.072	0.092	0.102	0.104	0.84	liberal
`us-KR`	0.058	0.052	0.062	0.066	0.056	0.94	slightly liberal
`us-CR2`	0.086	0.070	0.092	0.100	0.088	0.83	liberal
`BL+cs-KR`	0.077	0.038	0.052	0.070	0.158	0.73	liberal

Analysis	Mean	Min	Max, CS	Max, C07 Hybrid	Max, C07 CO	Min κ	Class
`BL+cs-CR2`	0.048	0.036	0.052	0.050	0.054	0.97	nominal
`BL+us-Satt`	0.087	0.072	0.094	0.092	0.094	0.86	liberal
`BL+us-KR`	0.059	0.050	0.068	0.060	0.066	0.96	slightly liberal
`BL+us-CR2`	0.080	0.068	0.086	0.090	0.086	0.86	liberal

`cs-Satt` is the published analysis. All figures are verified (`09-covariance-study-tables.R`). The visit-factor and spline versions fall in the same classes, with mean sizes within 0.005 of the linear versions. `us-KR` is slightly liberal by about 0.008, a little over 2 MCSE of the pooled rate. The excess appears in every `us-KR` variant, but they are fitted to the same trials and are not independent confirmations.

4.2 Why Kenward-Roger and CR2 are not interchangeable

Kenward-Roger [2] makes two adjustments. It inflates the model-based covariance of the fixed effects for the uncertainty in the estimated covariance parameters, and it matches the denominator degrees of freedom to the moments of the resulting Wald statistic. Both adjustments assume the working covariance is correct.

- **`us` is correct by construction.** Its only failure is the cost of estimating 45 parameters from 70 participants. That is what Kenward-Roger repairs: κ rises from 0.84-0.90 to 0.94-1.01, and the mean size falls from 0.090 to 0.058.
- **`cs` and `csh` under AR(1) data are misspecified.** Their failure is bias in the model-based variance, which Kenward-Roger does not address: κ stays at 0.73-0.75 in CO.

CR2 [3] replaces the model-based variance with a sandwich whose residuals are adjusted by the working model's hat matrix. It is consistent whatever the working covariance.

- **With a parsimonious working model** (2 to 3 parameters), the weights are nearly fixed and the bias reduction works. κ is 0.92-1.04, and every such analysis is nominal.
- **With `us`, the weights are themselves estimated from 45 parameters.** They adapt to the noise in each replicate, and the sandwich does not account for this. κ is 0.83-0.89, the same as the uncorrected model-based variance (an inferred mechanism; the κ values are verified).

The practical rule is to pair `us` with Kenward-Roger and a parsimonious working covariance with CR2.

4.3 Why the published model fails in CO under AR(1) data

The CO contrast compares two blocks of four visits, 2.5 weeks apart within a block, whose centers are ten weeks apart. Under AR(1) at $\rho = 0.7$ per week, the average correlation within a block is 0.272 and between blocks 0.069.

A compound-symmetry fit estimates one common correlation, the average over all 28 pairs, 0.156. For the difference of block means, in units of the per-visit variance:

- the true variance is $2(1 + 3 \times 0.272)/4 - 2 \times 0.069 = 0.770$;
- the compound-symmetry variance is $2(1 - 0.156)/4 = 0.422$.

The implied κ is $\sqrt{0.422/0.770} = 0.740$, against the observed 0.73 to 0.75 (derived; approximate, because it ignores the baseline row, the time term and the cross-component structure). In Hybrid most off-drug visits are one week from an on-drug visit, so the averaging does less harm, and κ is 0.92 to 0.97.

The failure therefore depends on the design. It would appear in any design that contrasts long, well-separated blocks under serial correlation.

4.4 The time adjustment

`docs/40`, `docs/42` and `docs/43` reported an unexplained excess size, about 0.07, for flexible time adjustments in Hybrid under carryover.

Table 4: Null rejection rate in the CS Hybrid cell at $t_{1/2} = 1$ by time adjustment (500 replicates, MCSE 0.0097)

Analysis	Linear	Spline	Visit
`cs-Satt`	0.054	0.070	0.072
`us-KR`	0.056	0.068	0.072
`BL+us-KR`	0.056	0.060	0.064
`cs-CR2`	0.058		0.058
`csh-CR2`	0.060		0.060

The excess is confined to this one cell. It is 2.0 to 2.3 MCSE above nominal and absent under CR2. Averaged over the eight cells, the visit factor and the spline have the same mean size as linear time in every analysis (0.058 for `us-KR`). The evidence for a systematic effect of the time form on size is therefore weak. Linear time is never worse in any cell, and it is the recommended form.

Correction (2026-10-07). Calling this evidence weak is wrong. The excess in this cell under model-based and Kenward-Roger inference was found independently in `docs/42` (Section 4 table: 0.075 spline and 0.080 visit factor against 0.056 linear after the 2026-10-09 anchored-carryover update, 0.073, 0.080 and 0.058 before; MCSE about 0.007), on different simulated trials, and is reproduced here (0.068 to 0.072). Two independent runs at 2 to 4 MCSE each make it a real size excess for model-based and Kenward-Roger inference in Hybrid under carryover, not Monte Carlo noise (inspected, `docs/42`; verified, `summary-alt250-null500-combined.csv`). It is absent under CR2. The recommendation of linear time stands, and is strengthened. The corresponding row of Section 10 should be read with this note.

5. Power among the correctly sized analyses

Size-adjusted power, linear time, in Hybrid (verified):

Table 5: Size-adjusted power of correctly sized analyses in the Hybrid design

Analysis	CS, $t_{1/2} = 0$	CS, $t_{1/2} = 1$	C07, $t_{1/2} = 0$	C07, $t_{1/2} = 1$
`us-KR`	0.688	0.424	0.940	0.756
`us-KR-Dbc0.5`	0.700	0.472	0.908	0.812
`RI+CAR1-CR2`	0.616	0.456	0.940	0.692
`cs-CR2`	0.664	0.428	0.828	0.676
`csh-CR2`	0.692	0.392	0.840	0.712
`sp_exp-CR2`	0.476	0.332	0.936	0.664
`BL+us-KR`	0.704	0.404	0.784	0.464
`BL+cs-CR2`	0.724	0.476	0.760	0.468
`cs-Satt` (published)	0.664	0.456	0.844	0.676

And in CO:

Table 6: Size-adjusted power of correctly sized analyses in the CO design

Analysis	CS, $t_{1/2} = 0$	CS, $t_{1/2} = 1$	C07, $t_{1/2} = 0$	C07, $t_{1/2} = 1$
`us-KR`	0.720	0.748	0.804	0.768
`us-KR-Dbc0.5`	0.716	0.748	0.796	0.776
`RI+CAR1-CR2`	0.684	0.704	0.688	0.732
`cs-CR2`	0.712	0.724	0.592	0.640
`csh-CR2`	0.688	0.688	0.608	0.688
`sp_exp-CR2`	0.336	0.312	0.660	0.668
`BL+us-KR`	0.708	0.756	0.492	0.520
`BL+cs-CR2`	0.784	0.792	0.504	0.524

The size-adjusted figures rest on a critical value estimated from 500 null replicates. That adds about 0.02 to their uncertainty beyond the power MCSE of up to 0.032, so differences under about 0.05 should not be read as real.

- **Under CS data the choice hardly matters,** with two exceptions. `sp_exp` loses a third of the power in Hybrid and half in CO, because it imposes a decay the data do not have. `RI+CAR1` loses about 0.07 in Hybrid.
- **Under AR(1) data the row-coded `us` analysis is the most powerful, or tied for it, in every cell.** `RI+CAR1` and `sp_exp` match it without carryover in Hybrid but fall behind with carryover and in CO.
- **Kenward-Roger costs no power.** At a common size, `us-KR` and `us-Satt` agree to within 0.008 in every cell (verified). Kenward-Roger changes the standard error and degrees of freedom, not the estimate, so the ordering of the replicates barely changes. The higher raw power of `us-Satt` (for example 0.956 against 0.944) is its excess size.
- **The baseline-covariate analyses lose 0.15 to 0.31 under AR(1) data.** Section 6 explains why.

6. The baseline row and the shared biomarker effect

In the row coding, the baseline row and the post-baseline off-drug visits share one biomarker main effect β_B. In the covariate coding, baseline is conditioned on and the biomarker's association with it is not used. The follow-up pass added a single analysis to separate the two explanations for the row coding's power: `us-KR-sepbm`, the row coding with an extra term $B_i \cdot \mathbf{1}\{t = 0\}$ that frees the biomarker effect at baseline.

Table 7: Raw power of `us-KR` with shared versus freed baseline biomarker effect (visit-factor time adjustment, 250 replicates per cell; columns give construct, design and $t_{1/2}$)

Analysis	CS Hybrid 0	CS Hybrid 1	C07 Hybrid 0	C07 Hybrid 1	C07 CO 0	C07 CO 1
`us-KR`	0.704	0.468	0.944	0.796	0.800	0.788
`us-KR-sepbm`	0.720	0.460	0.780	0.504	0.592	0.528
`BL+us-KR`	0.704	0.456	0.792	0.484	0.600	0.572

The figures are verified. `us-KR` shares β_B with the baseline row; `us-KR-sepbm` frees it.

Freeing β_B at baseline removes the whole advantage. The freed row model and the covariate model then agree to within Monte Carlo error in every cell. Their size is the same (0.058 and 0.059).

Correction to `docs/41`. Section 7 of `docs/41` derived that, with complete data, an unstructured covariance and β_B free at baseline, keeping baseline in the response is equivalent to conditioning on it (compare Liu et al. [5]). That derivation is confirmed. Its table, however, applied the equivalence to the published row coding, which shares β_B. That application is wrong: the shared form is a different, more constrained model, and under AR(1) data it is substantially more powerful.

What the constraint buys. In Hybrid, the post-baseline off-drug visits are few and late (two or three blinded-discontinuation visits and one crossover visit). The interaction is the difference between the biomarker's on-drug and off-drug associations. With the constraint, the baseline row adds a further, well-measured off-drug observation of the biomarker association. Under CS the participant effect absorbs most of what this adds, which is consistent with Proposition 4 of `docs/41`. Under AR(1) it does not. The mechanism is verified empirically by the `sepbm` arm; an analytic account has not been derived.

What the constraint assumes. The constancy assumption holds in both constructs by design: the biomarker is coupled only to the drug response, never to the baseline or to the natural-course and expectancy components. The null cells therefore cannot reveal what happens when it fails.

In a prazosin trial it fails if blood pressure is prognostic for the symptom course off drug: for example, if participants with higher pressure improve more, or less, on placebo or with time. The shared β_B would then be a compromise between two different associations. The interaction estimate would absorb the difference, and its null distribution would no longer be centered at zero (inferred, not simulated).

7. Carryover in the analysis: $D_{bc,it}$

`docs/43` asked whether the analysis should model carryover. The exposure pass answers this for `us-KR`, `us-CR2` and `RI+CAR1-CR2`.

Size. $D_{bc,it}$ does not change the size of any analysis. `us-KR` is 0.058 with D_{it}, and 0.057, 0.055 and 0.059 at $t^a_{1/2} = 0.5$, 1 and 2. `RI+CAR1-CR2` is 0.051 to 0.054 throughout. `us-CR2` stays liberal (0.085) (verified).

Power (`us-KR`, linear time, raw; verified):

Table 8: Raw `us-KR` power by exposure coding and assumed half-life $t^a_{1/2}$

Exposure coding	CS Hybrid 0	CS Hybrid 1	C07 Hybrid 0	C07 Hybrid 1	CO, all cells
D_{it}	0.700	0.436	0.944	0.812	0.752-0.808

Exposure coding	CS Hybrid 0	CS Hybrid 1	C07 Hybrid 0	C07 Hybrid 1	CO, all cells
$D_{bc,it}$, $t^a_{1/2} = 0.5$	0.708	0.492	0.928	0.852	0.748-0.808
$D_{bc,it}$, $t^a_{1/2} = 1$	0.632	0.504	0.892	0.884	0.744-0.808
$D_{bc,it}$, $t^a_{1/2} = 2$	0.564	0.468	0.840	0.876	0.736-0.796

- **In Hybrid it is a trade-off.** An assumed half-life longer than the true one spreads the contrast onto off-drug visits that carry no effect. One shorter than the true one leaves residual effect in the off-drug reference.
- **At $t^a_{1/2} = 0.5$ the trade-off is favorable in every cell.** It costs 0.016 or less without carryover and gains 0.04 to 0.06 at $t_{1/2} = 1$. The size-adjusted figures agree (Section 5).
- **In CO it makes little difference** (0.736 to 0.808 in every cell and at every assumed half-life). The first off-drug visit is 2.5 weeks after discontinuation, where $D_{bc,it}$ is 0.03 at $t^a_{1/2} = 0.5$ but 0.42 at $t^a_{1/2} = 2$. Why even the longer assumptions cost nothing in CO was not analyzed.

The estimand. With D_{it}, carryover attenuates the estimate, for example from -0.135 to -0.100 in CS Hybrid at $t_{1/2} = 1$. $D_{bc,it}$ with $t^a_{1/2} = t_{1/2}$ recovers the full on-drug interaction (-0.129). A longer assumed half-life overshoots it (-0.151 at $t^a_{1/2} = 2$; -0.168 without carryover). Under $D_{bc,it}$, $\beta_{bm:D}$ is the interaction per unit of modeled exposure, so its scale depends on $t^a_{1/2}$. The assumed half-life must be fixed, not chosen from the data, and the estimate is interpretable as the on-drug interaction only if the assumption is close to the truth.

For prazosin. The pharmacokinetic half-life of prazosin is a few hours, so a pharmacological carryover half-life of 0.5 week is generous. Carryover much longer than that would have to be behavioral or psychological (inferred). An assumed half-life of 0.5 week therefore covers the plausible pharmacological range, and costs almost nothing if there is no carryover.

`RI+CAR1-CR2` **with** $D_{bc,it}$ shows the same pattern at lower power, with a steeper cost at long assumed half-lives (0.412 at $t^a_{1/2} = 2$ in CS Hybrid without carryover, against 0.616 with D_{it}).

8. The pre-exposure coding of $D_{bc,it}$ in manuscript 02

Manuscript 02 builds the exposure-weighted indicator as 1 on drug and `carryover_decay(tsd, t_half)` off drug (`analysis/scripts/carryover-sensitivity/simulation-core.R`

lines 179 to 183; the same pattern at lines 662 and 688). Two facts combine (inspected):

- the package sets $t_{sd} = 0$ before the first exposure (`R/buildtrialdesign.R`, line 105);
- `carryover_decay(0, h)` returns $e^0 = 1$ for any half-life `h > 0` (`implementations/tidyverse/R/functions.R`, line 56, which the manuscript 02 driver uses; the `nof1power` copy behaves the same).

Off-drug visits that precede any exposure are therefore coded as fully exposed. In Hybrid every path starts on drug, so the defect does not arise; manuscript 02 also has no baseline row. In CO the placebo-first path is off drug for its first four visits, and these are coded 1. The indicator is then 1 at every visit for half the participants, and they contribute nothing to the within-participant contrast.

Check. `10-dbc-preexposure-check.R` fits the same trials with both codings, using manuscript 02's model (random intercept, `corCAR1`, no baseline row, linear time) at $t_{1/2} = t^a_{1/2} = 1$, with 250 replicates per cell (verified):

Table 9: Power (CR2 standard errors) under corrected and manuscript 02 pre-exposure coding of $D_{bc,it}$; the last column counts participants whose indicator is constant over all visits under manuscript 02's coding

Construct, design	D_{it}	$D_{bc,it}$ corrected	$D_{bc,it}$ as in 02	Constant indicator
CS, Hybrid	0.424	0.496	0.496	0
CS, CO	0.744	0.744	0.484	35
C07, Hybrid	0.476	0.572	0.572	0
C07, CO	0.548	0.552	0.256	35

- **In Hybrid** the two codings give identical fits, as they must.
- **In CO** the defect costs 0.26 to 0.30 in power, and it raises the empirical standard deviation of the estimate by a factor of 1.44 to 1.47 (from 0.049 to 0.070 under CS). That is close to the $\sqrt{2}$ expected from losing half the participants' contrasts.
- **The corrected coding performs like D_{it} in CO,** as in the covariance study.

Manuscript 02 reports exposure-weighted power of 0.488 against 0.830 for the binary coding in CO (its Section 3.4). It attributes this to "the continuous decayed predictor's mismatch with CO's design". The size of the loss matches the defect, and the corrected coding shows no mismatch. The reported finding is very likely an artifact of the coding (inferred: manuscript 02's own pipeline has not been rerun with the correction).

The same applies to its finding that AIC selection of the half-life picks the largest candidate in CO. That selection was made among models all affected by the defect.

Fix. Code pre-exposure off-drug visits as 0, for example `Dbc = ifelse(Db == 1, 1, ifelse(tsd > 0, carryover_decay(tsd, h), 0))`. Then rerun the CO cells of manuscript 02, its blocks S7 to S10 in particular.

Note (2026-10-07). The driver now applies this fix through `exposure_weight()` (`simulation-core.R`, lines 147 to 156), so the line numbers cited above refer to the version inspected when this paper was written (inspected). Whether manuscript 02's CO cells have been rerun with the corrected coding was not established here.

9. Recommendation for a Hybrid trial of prazosin with blood pressure

9.1 Primary analysis

Superseded (2026-10-07). This recommendation is replaced by analysis F1 (`docs/49`, Section 7; `docs/46`, Amendment 2). The exploratory stress test showed that this model fails the strict size criterion (0.057 pooled, 0.087 at $N = 35$) and that its shared baseline biomarker effect rejects 0.195 to 0.416 under a natural-course or placebo-prognostic biomarker. The text below is kept as written.

An MMRM for all nine rows (baseline as a row at $t = 0$, coded off drug):

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_t w_t + \beta_D D_{bc,it} + \beta_{bm:D} B_i D_{bc,it} + \varepsilon_{it}, \qquad \mathrm{Var}(\varepsilon_i) = \Sigma \text{ unstructured},$$

with B_i centered at its sample mean, $D_{bc,it}$ built at $t^a_{1/2} = 0.5$ week with pre-exposure visits coded 0, REML estimation, and Kenward-Roger inference. The test is the two-sided Wald test of $\beta_{bm:D} = 0$ at $\alpha = 0.05$.

```
library(mmrm)
dat$Dbc <- ifelse(dat$Db == 1, 1,
                  ifelse(dat$tsd > 0, 0.5^(dat$tsd / 0.5), 0))
dat$bmc <- dat$bm - mean(dat$bm[dat$t == 0])
fit <- mmrm(Sx ~ bmc + t + Dbc + bmc:Dbc + us(visit | ptID),
            data = dat, method = 'Kenward-Roger',
            vcov = 'Kenward-Roger-Linear')
summary(fit)$coefficients['bmc:Dbc', ]
```

Why this analysis.

- **Size:** slightly liberal, 0.057 on average, never above 0.064 in any cell with linear time.
- **Robustness:** its validity does not depend on the serial correlation structure.
- **Power:** it is the most powerful correctly sized analysis under AR(1) data and as powerful as any under CS data.
- **Carryover:** at $t^a_{1/2} = 0.5$ it insures against pharmacological carryover at almost no cost.

What the protocol must state.

- the assumed half-life and its justification;
- the constancy assumption (Section 6), with the reason to believe it;
- linear time as the time adjustment;
- that the unstructured covariance is estimated over all nine rows;
- the handling of missing visits: MMRM under missing at random, which this study did not test (Section 11).

9.2 Sensitivity analyses

1. β_B **freed at baseline.** Add `bmc:bl0`, where `bl0` is 1 at baseline. Equivalently, analyze the eight post-baseline visits with baseline as a covariate (`BL+us-KR`). Its size is the same; it gives up the constancy assumption at a power cost of up to 0.15 under AR(1) data. A material disagreement between the primary and this analysis is evidence against the constancy assumption.
2. `RI+CAR1-CR2` **with the same** $D_{bc,it}$**.** A nominally sized analysis on a different basis of inference (a robust variance about a parsimonious working model). Agreement shows the conclusion does not rest on the unstructured covariance.
3. D_{it} **in place of** $D_{bc,it}$**.** The published exposure coding, for comparability with Hendrickson et al. [1].

9.3 When to change the primary analysis

- **If the constancy assumption cannot be defended:** for example, if pilot data suggest blood pressure predicts placebo response or the natural course. Make sensitivity analysis 1 primary.
- **If the sample is much smaller than 70,** or visits are often missed, so that 45 covariance parameters are poorly determined: use `csh-CR2` or `RI+CAR1-CR2` as primary. Both are nominal in every cell, at a size-adjusted power cost of up to about 0.20 against the recommended primary under AR(1) data (least in Hybrid without carryover, most in CO). This study did not test smaller samples.

- **If the design is CO rather than Hybrid,** the same recommendation holds, and $D_{bc,it}$ makes no difference.

9.4 What not to use

- **The published random-intercept model with model-based inference (`cs-Satt`, equivalently `lmer`).** It is anticonservative under serial correlation in CO, 0.15 to 0.16.
- **`us` without Kenward-Roger, and `us` with CR2.** Both are liberal, 0.080 to 0.090.
- **`cs` or `csh` with Kenward-Roger.** Same failure as the published model.
- **`sp_exp` with model-based inference.** It is conservative and loses a third of the power under CS data.
- **Manuscript 02's $D_{bc,it}$ coding** in any design with off-drug visits before the first exposure.
- **A long assumed half-life chosen for safety.** It costs up to 0.14 in power when carryover is short.

10. Corrections to earlier documents

Table 10: Corrections to statements in earlier documents

Document	Statement	Correction
`docs/41`, Section 7 table	cLDA and ANCOVA coincide for the unstructured row model	True only with β_B free at baseline; the published row coding shares it and differs (Section 6)
`docs/41`, Section 8	Recommends baseline as a covariate for the trial analysis	Superseded by Section 9: the covariate form is the sensitivity analysis, or the primary if constancy cannot be defended
`docs/42` and `docs/43`, Section 4	The flexible-time size excess persists under every covariance and its cause is unexplained	Confined to one cell, absent under CR2 (Section 4.4); the excess under model-based and KR inference is real (2026-10-07 note, Section 4.4)
`docs/43`, Sections 1 and 5	`us-KR` is sized at 0.046 to 0.072	Slightly liberal, mean 0.058; the CR2 analyses with parsimonious covariances are nominal (Section 4)
`docs/43`, Section 6	The follow-up questions are open	Answered here (Sections 4 to 7)

Document	Statement	Correction
Manuscript 02, Section 3.4 and blocks S7 to S10	The exposure-weighted analysis collapses under CO	Likely an artifact of the pre-exposure coding (Section 8)

11. Limitations and evidence status

- **Two constructs, both satisfying the constancy assumption.** No null cell has a prognostic biomarker, so the row coding's size when the assumption fails is unknown. This is the most important open question for the recommendation. (Answered 2026-10-07 in `docs/46` and `docs/48`: under a natural-course or placebo-prognostic biomarker the row coding is anticonservative, up to 0.416 for `us-KR`.)
- **One sample size, no missing data.** $N = 70$ and complete data throughout. Kenward-Roger with `us` is known to degrade in small samples, and missing visits change both the covariance estimation and the cLDA-ANCOVA comparison (`docs/41`, Section 7).
- **Two half-lives.** $t_{1/2} \in \{0, 1\}$ week; the published values of 0.1 and 0.2 lie between them. The claim that $D_{bc,it}$ at 0.5 costs almost nothing at short carryover is interpolated.
- **Untested combinations.** $D_{bc,it}$ was crossed only with row-coded `us-KR`, `us-CR2` and `RI+CAR1-CR2`. Its size under the freed (β_B at baseline) and covariate codings is inferred from its invariance in the row coding. The expectancy term, random slopes and GEE were not part of this study (`docs/43`).
- **Monte Carlo precision.** Power MCSE is up to 0.032 per cell. Size MCSE is 0.0097 per cell and 0.0034 pooled. Classes near a boundary (`us-KR` at 0.058) could change with more replicates.
- **Mechanisms.** The account of the CR2 failure with `us` is inferred. The CS-under-AR(1) account (Section 4.3) is an approximate derivation that agrees with the observed κ. The row-coding mechanism is verified empirically but not derived.
- **Manuscript 02.** The defect is verified by inspection and its effect by simulation in this construct; manuscript 02's own pipeline has not been rerun.

12. Open questions

1. **Size of the row coding under a prognostic biomarker.** Couple the biomarker to the natural-course or expectancy component in the null cells. This decides between the primary and sensitivity analysis 1. (Done in the

stress test of `docs/46`, cells 11 to 13 and 40 to 41; see `docs/48`, Section 2.)

2. **The recommended primary and sensitivity analyses in the remaining conditions.** Run them at $t_{1/2} \in \{0.1, 0.2, 0.5\}$, at $N \in \{30, 50\}$, and with the published dropout mechanisms.
3. **Manuscript 02 with the corrected coding.** Rerun its CO cells.
4. **An analytic account of the shared-β_B gain under AR(1),** along the lines of `docs/41`, Proposition 4.

13. Reproducibility

Run from the repository root:

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R \
  --reps-alt 250 --reps-null 500 --cores 8
Rscript analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R \
  --reps-alt 250 --reps-null 500 --cores 8 --arms followup
Rscript analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R \
  --reps-alt 250 --reps-null 500 --cores 8 --arms dbc
Rscript analysis/scripts/quick-sim/carryover-closed-form/09-covariance-study-tables.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/10-dbc-preexposure-check.R \
  --reps 250 --cores 4
```

- **Study passes (the first three commands).** Each writes `summary-alt250-null500{,-followup,-dbc}.csv` and the matching replicate files to `analysis/data/quick-sim/carryover-closed-form/covariance-study/`. Chunks are checkpointed under `cells-*`, so an interrupted pass resumes.
- **Tables (`09`).** Combines the passes into `summary-alt250-null500-combined.csv` and computes `size-adjusted-power-alt250-null500.csv`.
- **Pre-exposure check (`10`).** Writes to `analysis/data/quick-sim/carryover-closed-form/dbc-preex`
- **Run time.** On 8 cores the three passes took about 3, 25 and 23 hours of wall time. The last two include periods when the machine slept. The check took about 15 minutes.

14. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.
2. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics* 1997; 53(3):983-997. doi:10.2307/2533558.

3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology* 2002; 28(2):169-181. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics* 2018; 36(4):672-683. doi:10.1080/07350015.2016.1247004.
4. Sabanés Bové D, et al. *mmrm: Mixed Models for Repeated Measures*. R package, CRAN.
5. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine* 2009; 28(20):2509-2530. doi:10.1002/sim.3639.
6. pmsimstats team. `docs/37-carryover-power-closed-form.md`, `docs/40-hendrickson-hybrid-design-primer.md`, `docs/41-baseline-as-covariate.md`, `docs/42-flexible-time-adjustment.md`, `docs/43-analysis-approaches-catalog.md`, `docs/44-dgp-roadmap.md`.